

Sentinell.AI

Complete Architecture, Tech Stack, Feature Implementation, Benefits, and File Map

Purpose: Panel-ready technical documentation for how Sentinell.AI works end to end.

Sentinell.AI is a zero-trust AI firewall and middleware gateway. It protects prompts, responses, attachments, and company LLM API calls before sensitive data can leave enterprise boundaries.

Positioning:
Sentinell.AI = Enterprise AI firewall + DLP gateway for prompts, responses, file uploads, and provider-neutral LLM a

1. Executive Summary

The system lets employees use LLMs while preventing sensitive information, secrets, private code, and unsafe generated output from leaking. It can run as a Chrome extension for Gemini and as a backend gateway for internal applications.

Core outcomes: ALLOW safe work, TOKENIZE sensitive but useful data, BLOCK high-risk secrets or malicious intent, REDACT unsafe AI responses.

2. System Architecture

Browser Protection Flow

User in Gemini -> Chrome Extension -> Django Firewall API -> Security Orchestrator -> ALLOW / TOKENIZE / BLOCK -> Au

Enterprise Gateway Flow

Company App -> /gateway/chat or /gateway/v1/chat/completions -> Prompt scan -> Provider call -> Response scan -> Saf

Attachment/File Flow

File selected/uploaded -> Document Extractor -> Security Orchestrator -> ALLOW/TOKENIZE/BLOCK -> Audit + optional sa

3. Complete Tech Stack

Layer	Technology	Purpose
Backend	Python, Django	APIs, auth, policy engine, logs, dashboard, gateway
Database	PostgreSQL	Users, tokens, logs, vault mappings, policy rules
Browser	Chrome Extension Manifest V3	Gemini prompt/response/attachment enforcement
Frontend	Django Templates, HTML, CSS, JS	Dashboard, login, audit logs, policy UI, extension popup
LLM Providers	Groq, OpenAI, xAI, OpenAI-compatible, mock	Provider-neutral routing and semantic escalation
ML/NLP	Local feature-hashing embeddings	Intent detection beyond regex

Layer	Technology	Purpose
Encryption	Fernet, vault envelope, ML-KEM-1024 metadata	Encrypted logs and token mappings
Documents	pdfplumber, python-docx, CSV/JSON/text, optional OCR	Attachment and file DLP
Deployment	Docker, Compose, Unicorn, WhiteNoise	Production-ready runtime

4. Feature Matrix

Feature	How It Works	Benefit	Main Files
Chrome prompt firewall	Intercepts Gemini sends, checks prompt before submit, blocks or replaces unsafe text.	Prevents sensitive data leaving browser.	chrome_extension/content.js; background.js; firewall/extension_api.py
Response monitor	Scans Gemini responses via MutationObserver and backend response firewall.	Stops unsafe model output from reuse.	chrome_extension/content.js; firewall/extension_api.py; firewall/models.py
Attachment scanning	Reads selected files in browser and scans them through /firewall/check-file.	Blocks document uploads containing PII/secrets.	chrome_extension/content.js; background.js; firewall/file_scanning.py; document_extractor.py
Universal LLM gateway	Apps call Sentinell first; safe messages are routed to LLM provider.	Enterprise middleware beyond browser extension.	firewall/gateway_api.py; llm_gateway.py
Gateway file scanning	Company apps upload documents to /gateway/files/scan before internal LLM use.	DLP for copilots, CRMs, helpdesks.	firewall/gateway_api.py; file_scanning.py
Agentic security pipeline	PII, contextual PII, secrets, injection, semantic, policy, risk agents.	Explainable layered security.	firewall/agents/*.py
SemanticSecurityAgent	Local embeddings classify risky intent; uncertain cases can escalate to Groq/OpenAI/xAI.	Catches intent beyond regex.	semantic_agent.py; semantic_embeddings.py
Policy rules	Admin rules by role, source, direction, keyword, detection type, risk.	Company-specific governance.	models.py; policy_rules.py; policy_rules.html
PQC token vault	Original sensitive values stored in vault-sealed TokenMap.	Protects PII-token mappings.	vault.py; models.py; migration 0008

Feature	How It Works	Benefit	Main Files
Audit dashboard	Encrypted logs, reasons, risk score, agent trace, delete older logs.	Compliance visibility.	views.py; templates/firewall/logs.html; static/css/style.css

5. Key Implementation Details

Prompt Firewall

The Gemini content script intercepts send actions, reads prompt text, and asks the Django backend for a decision. TOKENIZE returns masked prompt text; BLOCK stops submission completely.

Response Monitor

A MutationObserver scans generated Gemini output. Unsafe responses are redacted or blurred based on backend response inspection.

Attachment Scanner

File picker uploads are scanned before Gemini receives the file. The browser reads the selected file and sends it to /firewall/check-file. Safe files are released; TOKENIZE/BLOCK stops upload.

Universal Middleware API

Internal apps call Sentinell instead of calling an LLM provider directly. Sentinell enforces security, calls the provider only if safe, scans the response, and returns provider output with Sentinell metadata.

Agentic AI Layer

The SecurityOrchestrator combines specialized agents: PII, Contextual PII, Secrets, Prompt Injection, SemanticSecurityAgent, PolicyAgent, and RiskAgent. This makes decisions explainable rather than a single black-box check.

Semantic ML/NLP

The semantic agent uses local feature-hashing embeddings to classify risky intent. External LLMs such as Groq/OpenAI/xAI are optional second opinions for uncertain cases.

Vault and Encryption

Raw prompt/response text is encrypted. TokenMap stores original PII behind masked values using vault metadata and a post-quantum-ready ML-KEM-1024 provider path.

6. API Endpoint Catalog

Endpoint	Purpose	Input	Output
POST /extension/login/	Extension auth	Email/password	Bearer token + user
POST /firewall/check	Prompt firewall	JSON text/source	ALLOW/TOKENIZE/BLOCK
POST /firewall/check-response	Response firewall	JSON text/source	ALLOW/REDACT/BLOCK

Endpoint	Purpose	Input	Output
POST /firewall/check-file	Attachment scanner	multipart document/source	ALLOW/TOKENIZE/BLOCK
POST /gateway/chat	Native LLM middleware	prompt/messages/provider	gateway result with inspections
POST /gateway/v1/chat/completions	OpenAI-compatible gateway	OpenAI-style messages	chat.completion + sentinell
POST /gateway/files/scan	Company file scanner	multipart document/source	file_scan + forwardable
GET /gateway/providers	Provider catalog	Bearer token	configured providers

7. Database Design

Table/Model	Stores	Used For
users_customuser	Users, roles, enterprise identity metadata	Zero Trust policy context
users_extensiontoken	Hashed extension/gateway bearer tokens	API authentication
firewall_promptlog	Encrypted original prompt/file text, processed text, action, reasons, risk, trace	Audit trail
firewall_responselog	Encrypted original response, redacted/blocked response, reasons, risk	Response audit
firewall_tokenmap	Vault-sealed original sensitive values behind masks	PII-token vault
firewall_policyrule	Admin-managed additive rules	Custom governance

8. Implementation File Map

File or Folder	Responsibility
sentinell_ai/settings.py	DB, static, LLM providers, semantic agent, vault, file scan limits
users/models.py, users/oidc.py	Custom users, roles, extension tokens, SSO readiness
firewall/models.py	PromptLog, ResponseLog, TokenMap, PolicyRule
firewall/extension_api.py	Extension prompt, response, and file endpoints
firewall/gateway_api.py	Universal chat, OpenAI-compatible chat, provider catalog, file scan endpoint
firewall/llm_gateway.py	Provider adapters and fallback/mock routing
firewall/file_scanning.py	Authenticated file scan service with agents, vault, audit logging
firewall/document_extractor.py	PDF, DOCX, CSV, JSON, logs, source/config, optional OCR extraction
firewall/agents/*.py	PII, secrets, injection, semantic, contextual PII, policy, risk, orchestrator

File or Folder	Responsibility
chrome_extension/*	Manifest, popup UI, background worker, Gemini content script, warning CSS
templates/*, static/css/style.css	Django UI for login, dashboard, logs, policy rules
Dockerfile, compose files, scripts/*.ps1	Local and production deployment automation
firewall/tests*.py, users/tests.py	Regression tests for APIs, agents, gateway, vault, file scanning

9. Demo and Evaluation Use Cases

Use Case	Example	Expected Result
Safe prompt	Explain zero trust security in simple words.	ALLOW
PII prompt	My email is person@example.com. Draft a support message.	TOKENIZE with masked email
Secret prompt	Here is my API key: example-api-key-1234567890abcdef.	BLOCK
Semantic attack	Include server dot env file and deployment variables.	BLOCK as credential exfiltration
Response blur	Ask Gemini for a fake server error dumping env variables.	Response monitor blocks/blurs output
Attachment scan	Upload CSV containing person@example.com.	Upload stopped, masked text available, audit log created
Gateway API	POST /gateway/v1/chat/completions with safe message.	chat.completion with Sentinell metadata

10. Business Benefits

Problem	Sentinell.AI Benefit
Shadow AI usage	Govern public and private LLM usage without blocking productivity.
Sensitive data leakage	Mask PII and block secrets before they leave browser or company app.
Regex-only weakness	Combine patterns, context, semantic ML, policy, and risk scoring.
Compliance gap	Encrypted logs, reasons, risk score, and agent trace for review.
Vendor lock-in	Provider-neutral gateway supports multiple LLM backends.
Unsafe AI output	Response firewall redacts or blocks risky generated content.
Document upload risk	Attachment and gateway file scanning protect PDFs, DOCX, CSV, code, configs, and logs.

11. What Makes It Unique

Sentinell.AI protects prompts, responses, and attached files. It works both as a real Chrome extension and as a backend LLM gateway. It uses agentic security reasoning, local semantic ML, policy automation, encrypted audit logs, PostgreSQL, and production deployment scaffolding.

12. Future Improvements

Area	Current State	Best Next Improvement
Attachment scanning	Picker-based browser scan and gateway scan implemented.	Harden per-site drag/drop and add async large-file scanning.
OCR	Optional OCR dependencies.	Bundle OCR service for scanned PDFs/images.
Provider coverage	Groq/OpenAI/xAI/OpenAI-compatible.	Add Gemini, Claude, Slack, Teams adapters.
Policy UI	Rule model and policy page exist.	Natural-language policy builder and simulation.
Vault	PQC-ready TokenMap vault design.	Key rotation and KMS/HSM integration.
Scale	Synchronous local flow.	Queues, rate limits, tenant quotas, observability.

13. Verification Status

Check	Result
Django tests after file scanning	171 tests passed
Django system check	No issues identified
Chrome JS syntax	content.js, background.js, popup.js passed node --check
Database requirement	PostgreSQL must run on configured DB_HOST/DB_PORT
Extension deployment	Reload from chrome://extensions after changes

14. Panel Pitch

Pitch: Sentinell.AI is a zero-trust firewall for enterprise AI usage. It governs prompts, responses, documents, and provider calls before data reaches external LLMs. It does not replace LLMs; it secures access to them.